



Windows 8 on Chromebook

Watch this video showing the Chromebook Pixel running Windows 8 remotely using the Chrome Remote Desktop App. <http://dgit.in/YWjyFZ>

Triple storage with Google

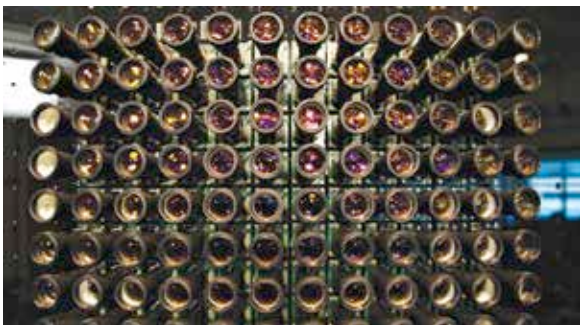
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it that Apple, HTC and even Nokia have been experimenting with the technology and could implement it into their devices in the coming years.

Super Sensors to the Rescue

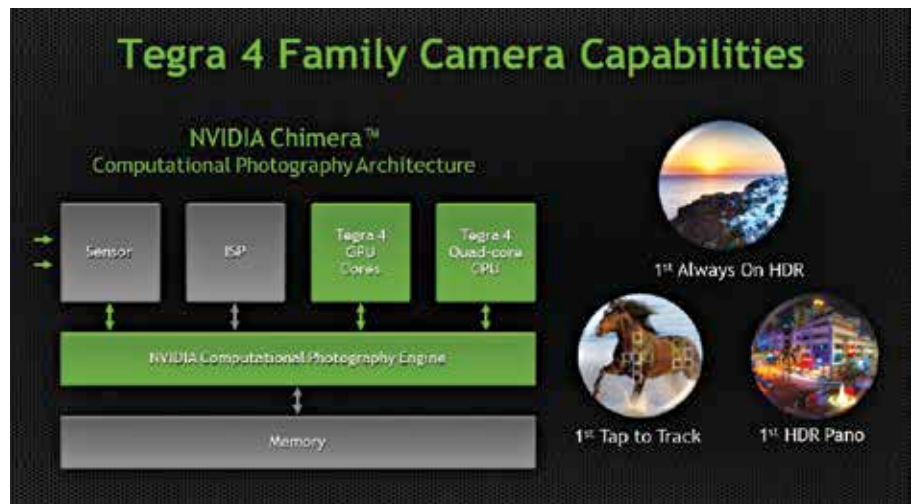
Lenses may be part of the magic, but EVERYTHING rests on the sensor's ability to capture the intended subject matter. It's not just about the silicon, but also about how individual photosites are arranged (think Bayer pattern sensors vs. Fuji's X-Trans sensors), how the wiring matrix is designed (think regular CMOS sensors vs. Back-Side Illuminated CMOS Sensors) and how the micro lenses filter the light coming in. Each of these properties has just recently begun to get refined, with the micro-lens array being the primary focus at the moment, with Panasonic doing some major ground breaking work in the field.

At the moment, a conventional imaging sensor employs micro lenses that only let either red, blue or green wavelength through to the respective photosite on the sensor. What this results in is the sensor getting only a fraction of the light, meaning for a particular colour to register, there must be a lot of light. Panasonic's new "micro colour



Stanford prototype Plenoptic Light Field camera array

splitters" are micro lenses which are able to separate the three primary colours and redirect them to their respective photosites using diffraction. The result is extreme sensitivity to colour information even in ridiculously low light, since none of the incoming light is being wasted. While Panasonic has a working prototype of the sensor ready, they say it is currently at the laboratory stage and still a few years away from finding its way into your camera. However, we wouldn't be surprised to see this technology becomes mainstream in the next five years.



While Panasonic is working hard to make the sensor far more efficient within the existing parameters (the micro colour splitters can be manufactured using existing technology) NVIDIA is developing its own architecture known as Chimera which is expected to be a mean little sensor. Taking advantage of the Tegra 4 processor which NVIDIA hopes to infuse many of the future smartphones with, the Chimera architecture will utilize the CPU and GPU capabilities of

Tegra 4 to generate real time HDR photos and videos. As of now, HDR video is only possible through the incredibly high end RED EPIC (which costs close to \$20,000) and even there, it is only in experimental stages. Imagine being able to shoot HDR video AND stills simultaneously using the Chimera architecture.

Game of Optics

There's a lot of sensor talk, but what about optics? Nokia has already revolutionized this arena with their first ever Optically Stabilized lens arrangement in the Lumia 920, but that's just one of the many innovations. What about the ability to change lenses on your camera phone, just like in a DSLR. The idea isn't so farfetched if you think about it. Currently, camera phones utilize a fixed lens design to ensure a slim form factor. If this was taken out of the equation, it could be possible to design the sensor assembly

in such a way so as to accept different kind of lenses. It could be something as simple as a "drop-in" lens block. Companies like Olloclip are already making lenses that can be attached to the front of the camera assembly on various smartphones, and while the results are pretty neat, there is some obvious loss in quality to do a not-so-optimized optical design. Being able to just replace the stock lens with a "telephoto" block or a "fish-eye"

The Endless Possibilities

As of now, we're seeing real development on the actual sensor, with light-field architecture being one of the main points of focus, while NVIDIA is trying to beef up the processor to a point where it can do far more in one go than previously imaginable. But what if one day, the light-field guys and the NVIDIA guys sit down at some café in South of Florence and decide to let their powers combine? Imagine having a camera phone with a light-field sensor, capable of shooting HDR video and HDR stills simultaneously. The possibility of being able to swap out the stock lens on your camera phone for a special macro lens or a telephoto lens makes the prospects of cellphone photography seem so much more promising. What's great is that we are seeing some "primitive" form of all these technologies be developed today and at the rate at which science performs miracles, it won't be long before your cellphone replaces your regular point and shoot for good. 